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WEEK THREE ANSWER SHEET

One important hardware component in a network is a **router**, while a key software component is a **network operating system (NOS)**. A router is responsible for directing data packets between different networks, ensuring that information reaches the correct destination efficiently. For example, in a home or school network, the router connects multiple devices (phones, laptops, printers) to the internet and manages traffic so that each device can communicate without interference.

It improves performance by selecting the best path for data transmission and can enhance security through features like firewalls and access control. On the other hand, a network operating system (such as Windows Server or Linux-based systems) manages network resources and enables communication between devices. It controls user access, monitors network activity, and ensures secure data sharing. For instance, in a school network, the NOS allows students and teachers to log in, access shared files, and use printers while maintaining data security and system efficiency.

In managed networks, one of the most important features for ensuring reliable communication is **network monitoring**. This feature allows administrators to continuously observe network performance, detect faults, and respond quickly to issues before they disrupt users. For example, in a school or organization, network monitoring tools can identify slow connections, unauthorized access attempts, or device failures in real time.

This ensures that corrective actions, such as fixing a faulty switch or blocking suspicious traffic, are taken immediately. Compared to unmanaged networks, where problems may go unnoticed until they become serious, managed networks provide better reliability, security, and performance. Therefore, network monitoring is crucial because it helps maintain smooth communication, minimizes downtime, and ensures that the network operates efficiently at all times.